

CONSTRUCTION OS

PRD - Client & Supplier Portals

Implementation-ready Product Requirements Document for Product, UX/UI, Frontend, Backend, Platform, Security, QA, DevOps and Operations Teams

Product purpose: Create secure, relationship-based external workspaces that expose only approved ConstructionOS information and convert every client, supplier and subcontractor action into a controlled, traceable submission.

Document control	Value
Document	PRD_Client_Supplier_Portals
Version	1.0
Status	Implementation-ready functional and technical specification
Audience	Product, UX/UI, frontend, backend, platform, security, QA, DevOps, SRE, client services, procurement and project operations teams
Primary routes	/client, /partner, /portal/admin, /portal/invitations, /portal/document-rooms
Core architecture	Shared external identity and access foundation + separate client and supplier experiences + permission-safe projections + immutable external submissions
Access model	Deny-by-default, relationship-based, tenant/company/project/unit/contract/package scoped, role and capability controlled, effective-dated and fully auditable
Source-of-truth policy	Internal modules own business truth; portal projections control visibility; external actions never directly overwrite authoritative internal records
Initial integration policy	Self-contained initial release. External e-signature, payment gateways, identity providers and third-party storage remain future roadmap integrations
Platform Configuration	Mandatory for portal type, page, widget, field, action, relationship, publication policy, branding, file policy, acknowledgment and retention
Date	2026-08-06

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1. Executive Summary

The Client & Supplier Portals Module is the controlled external collaboration layer of ConstructionOS. It provides purpose-built experiences for clients, buyers, owners, suppliers, subcontractors and consultants without exposing internal employee navigation, unrestricted search, internal comments, confidential pricing or operational records beyond the external party's active business relationships.

The module consists of a shared external identity, invitation, organization, membership, permission, document exchange, notification and audit foundation, combined with two distinct products: the Client Portal and the Supplier & Subcontractor Portal. Each portal uses permission-safe projections rather than unrestricted queries against internal modules. Every external upload, response, acknowledgement, quotation, claim, invoice, request or defect is preserved as an immutable external submission and only affects internal authoritative records after controlled internal processing.

Permanent rule: Internal modules own business truth. Portal projections control visibility. External users may view, submit, respond and acknowledge, but they never directly overwrite authoritative internal records.

1.1 Required outcomes

- Provide secure external access without issuing internal employee accounts.
- Limit visibility through explicit business relationships and effective-dated scopes.
- Offer dedicated, simplified client and supplier journeys rather than copies of internal screens.
- Preserve complete submission, document, acknowledgement, access and download history.
- Protect confidential internal fields, other external parties and unpublished records.
- Allow external collaboration across projects, units, contracts, procurement, finance, QA/QC, HSE, handover and warranty.
- Remain fully usable without third-party e-signature, payment, storage or identity integrations.

1.2 Primary implementation decision

Architecture decision: Build the portals on permission-safe, versioned projections and immutable external submissions. Do not allow portal clients to query unrestricted operational records or write directly to internal source entities.

2. Product Vision and Business Outcomes

2.1 Product vision

ConstructionOS should give every authorized external party one reliable place to access the information, documents, deadlines, financial statuses, submissions and actions relevant to its relationship with the business. The experience must feel premium and transparent while remaining intentionally narrower than the internal platform.

2.2 Business outcomes

- Reduce fragmented communication across email, messaging applications and uncontrolled file shares.
- Shorten response cycles for client decisions, supplier quotations, submittals, claims, invoices and corrective actions.
- Improve confidence through clear statuses, deadlines, issued records and acknowledgment evidence.
- Lower data-leakage risk by replacing ad hoc external sharing with relationship-based access.
- Improve document control and eliminate duplicate file copies through central Documents references.
- Provide scalable onboarding for external organizations and their authorized representatives.
- Increase client satisfaction and supplier collaboration without weakening internal governance.

2.3 Failure risks to prevent

Risk	Required prevention
Portal becomes an internal-app copy	Dedicated external IA, terminology, widgets, simplified flows and permission-safe projections.

Risk	Required prevention
External user sees another party's data	Organization, relationship, record and field isolation in APIs, indexes, caches and exports.
Submission overwrites internal truth	Immutable submission revision plus explicit internal acceptance/conversion.
Internal comments leak externally	Separate internal and external comment channels with explicit publication.
Document folder inheritance leaks files	External publication classification and document-room membership evaluated per item.
Account remains active after relationship ends	Account status separated from effective business access; automatic expiry and revocation.
Supplier bank details changed fraudulently	Sensitive-change workflow, reauthentication, verification and Finance-side acceptance.
Portal depends on third parties	In-app identity, documents, notifications and submissions operate without external integrations.

3. Success Metrics and Product Health

3.1 Product metrics

Metric	Initial success target / interpretation
Invitation activation rate	Percentage of valid invited users who activate within the invitation window.
Portal task completion	Percentage of required acknowledgements, submissions or responses completed before deadline.
External response time	Median time from publication/request to external response.
Document-control adoption	Share of external exchanges completed through controlled document rooms/transmittals.
Support burden	Portal support cases per 100 active users and repeated issue categories.
Submission return rate	Percentage returned for missing or invalid information; used to improve forms and guidance.
Security health	Unauthorized-access attempts, cross-organization denials, suspicious sessions and file-security failures.
Portal availability	Successful authenticated sessions and critical flow completion.

3.2 Guardrail metrics

- No performance score may be automatically assigned to a client, supplier or employee based only on portal activity.
- Login frequency and read rates are operational adoption indicators, not contractual performance evidence unless explicitly agreed.
- Portal analytics must not reveal restricted data through small-group aggregation.

4. Scope, Non-Goals and Assumptions

4.1 In scope

- Shared external identity, invitations, MFA, external organizations, memberships, roles and access relationships.

- Client portal for project, unit, progress, financial, contract, document, meeting, request, handover, defect and warranty views.
- Supplier portal for profile, qualification, RFQ, quotation, PO, delivery, submittal, claim, timesheet, invoice, QA/QC and HSE collaboration.
- External document rooms, controlled uploads/downloads, transmittals, required reading and acknowledgements.
- Permission-safe projections, immutable submissions, APIs, events, audit, responsive experiences and operational monitoring.
- Controlled BIM/3D viewing using approved model versions, layers and metadata.

4.2 Non-goals

- Giving external users internal employee accounts, departments, Teams membership or internal navigation.
- Replacing CRM, Projects, Units, Contracts, Finance, Procurement, Documents, QA/QC, HSE or BIM as source modules.
- Implementing the final universal Approvals & Workflow Engine.
- Providing unrestricted public project/model sharing.
- Implementing external e-signature, payment gateways, external identity providers or external storage in the initial release.
- Allowing external users to edit internal financial, project, supplier, contract or quality records directly.

4.3 Assumptions

- One external person may belong to multiple external organizations and relationships.
- One external organization may collaborate with multiple legal companies and projects.
- Access may differ by project, unit, contract, package and time period.
- External records and files may contain untrusted content and require quarantine/validation.
- Portal branding and terminology may vary by legal company while security remains uniform.

5. Terminology and Core Concepts

Term	Definition
External person	An individual who is not acting through an internal employee identity in the current portal context.
External organization	Client, supplier, subcontractor, consultant, investor or other non-internal legal/business entity.
Portal account	Authentication identity, sessions, devices, MFA and security state.
Portal membership	Effective-dated relationship between an external person and external organization.
Business access relationship	Explicit link between a membership/organization and allowed projects, units, contracts, packages or document rooms.
Portal projection	Permission-safe, externally worded representation of selected source-module data.
External submission	Immutable externally created record submitted for internal review or conversion.
Publication	Controlled act of making a projection or document visible to a defined external audience.
Acknowledgement	Evidence that a user received/read a specified item; not the same as formal approval.
Document room	Scoped external file workspace with explicit audience, actions, expiry and audit.

6. Permanent Source-of-Truth Boundaries

Domain	Source owner	Portal responsibility
Projects	Projects Module	Show approved project overview, milestones and external-safe activity.
Units / properties	Units Module	Show assigned unit data, selections, payment plan and handover status.
Clients	CRM	Resolve client organization/person relationships and external contact projections.
Suppliers	Suppliers / Procurement	Show supplier profile/qualification projections; collect proposed updates.
Contracts	Contracts	Show issued summaries, documents, obligations, notices and amendments.
Finance	Finance	Show permitted invoices, receipts, statements, claim/invoice/payment statuses.
Procurement	Procurement	Show RFQs, POs, package dates, clarifications, delivery and award outcomes.
Documents	Documents	Own files, versions, classifications, transmittals, retention and legal hold.
Progress	Work Progress	Provide client/supplier-safe progress projections and claim verification references.
Quality	QA/QC	Own inspection, NCR, defect and closure decisions.
Safety	HSE	Own eligibility, incidents, corrective actions and safety classifications.
BIM/3D	BIM/3D	Own approved models, versions, object identity and visual relationships.
Tasks	Tasks	Own internal execution created from external requests/submissions.
Notifications	Notifications & Event Centre	Own delivery channel, timing, digest, escalation and delivery history.

Boundary enforcement: Portal writes create submissions, requests, comments or acknowledgements. Internal source records change only through the owning module and its authorized process.

7. Personas and Stakeholders

Persona	Primary goals	Critical restrictions
Client executive	Understand progress, financial exposure, decisions and major risks.	No internal cost/margin, other clients, supplier bids or internal strategy.
Buyer / unit owner	View unit, payments, selections, handover, defects and warranty.	Only assigned unit/relationship and published records.
Client technical reviewer	Review approved designs, submissions, inspections and responses.	No unpublished drafts or internal comments.
Supplier administrator	Manage own organization users, profile and compliance.	Cannot grant access beyond assigned organizational scope.
Tender manager	Review RFQs, clarifications and submit quotations.	Cannot view competing suppliers or internal evaluation.

Persona	Primary goals	Critical restrictions
Supplier project/site manager	Manage deliveries, submittals, claims, QA/QC and HSE actions.	Only assigned projects/contracts/packages.
Supplier finance contact	Submit invoices and view payment statuses.	No internal journals, margins or other vendors.
Internal portal administrator	Configure portal policies, branding, templates and external access.	Does not bypass source-module permissions.
Internal relationship owner	Invite parties, publish records and review submissions.	Actions remain scoped and audited.

8. Portal Roles and Capability Model

8.1 Client roles

- Client Organization Administrator
- Client Executive
- Project Representative
- Buyer / Unit Owner
- Financial Contact
- Technical Reviewer
- Handover Representative
- Warranty Contact
- Read-only Viewer
- Custom Client Role

8.2 Supplier roles

- Supplier Organization Administrator
- Commercial Contact
- Tender Manager
- Contract Manager
- Project Manager
- Site Manager
- Document Controller
- Finance Contact
- Delivery Coordinator
- HSE Representative
- QA/QC Representative
- Read-only Viewer
- Custom Supplier Role

8.3 Capability model

Capability	Description
portal.view	View permitted published projections.
portal.download	Download permitted files/exports.
portal.upload	Upload files into an allowed submission/document room.
portal.submit	Submit a completed external record revision.
portal.comment	Create an external-visible comment in a permitted context.
portal.respond	Respond to clarification, notice or decision request.
portal.acknowledge	Record acknowledgement of a specific item/version.
portal.accept	Perform an explicitly configured legal/commercial acceptance action.

Capability	Description
portal.manage_members	Invite, suspend or manage organization members within delegated scope.
portal.export	Generate permitted external reports/exports.

Roles grant capabilities only inside explicit, effective-dated access scopes. A role never creates project, contract, package or document access by itself.

9. Product and Technical Architecture

9.1 Logical architecture

Internal Source Modules

- > Publication Policies / Source Adapters
- > Permission-Safe Portal Projections
- > Portal APIs
- > Client Portal / Supplier Portal

External User Actions

- > Immutable Submission / Request / Acknowledgement
- > Internal Review or Conversion
- > Source Module Update

9.2 Shared foundation and separate experiences

Shared services	Client-specific	Supplier-specific
External identity, invitations, MFA, sessions	Projects and progress	Prequalification and compliance
Organizations, memberships, access relationships	Units, payments and selections	RFQs, quotations and POs
Documents, messages, notifications, acknowledgements	Handover, defects and warranty	Deliveries, submittals and claims
Audit, branding, localization and support	Client decisions and variations	Invoices, QA/QC and HSE

9.3 Deployment posture

- Separate client and partner route spaces or hostnames with shared backend foundation.
- Independent portal release controls may be used while maintaining common identity and policy contracts.
- Permission-safe projections are cached independently from operational databases and invalidated by events.
- External file uploads are quarantined before becoming visible to internal source workflows.

10. Functional Requirements Register

ID	Area	Requirement	Priority	Acceptance signal
FR-001	Identity	Create a portal account without creating an internal employee account.	Must	Internal navigation and employee permissions are absent.
FR-002	Invitation	Issue single-use, expiring invitations tied to portal type, organization and access scope.	Must	Activation cannot exceed invitation scope.

ID	Area	Requirement	Priority	Acceptance signal
FR-003	MFA	Require configurable MFA and reauthentication for sensitive actions.	Must	Policy is enforced by portal and action.
FR-004	Membership	Allow one person to hold multiple external organization memberships.	Must	User switches contexts without cross-organization leakage.
FR-005	Relationships	Grant access only through explicit business access relationships.	Must	Active account with no relationship sees no business data.
FR-006	Expiry	Expire or revoke relationship access without deleting audit history.	Must	Previously available records disappear immediately.
FR-007	Projection	Serve external data only through permission-safe projections.	Must	Portal APIs never return unapproved fields.
FR-008	Publication	Support draft, published, withdrawn and expired external publication states.	Must	Only published/current projection is externally visible.
FR-009	Client project	Show approved project overview, progress, milestones and documents.	Must	All data is traceable to source records.
FR-010	Client unit	Show assigned unit details, selections, payments, handover and warranty.	Must	Other units remain inaccessible.
FR-011	Client finance	Show permitted invoice, receipt, payment-plan and balance projections.	Must	No internal accounting fields appear.
FR-012	Client requests	Allow client requests, variation proposals, document requests and inquiries.	Must	Each creates immutable submission history.
FR-013	Defects	Allow client defect/warranty evidence capture and status tracking.	Must	QA/QC retains formal acceptance/closure.
FR-014	Supplier profile	Allow supplier to propose profile and compliance updates.	Must	Supplier master changes only after internal acceptance.
FR-015	Prequalification	Support configurable prequalification questionnaire and evidence.	Must	Submission and review states are visible separately.
FR-016	RFQ	Expose RFQ scope, documents, deadlines, addenda and clarifications.	Must	Other suppliers and internal evaluation remain hidden.
FR-017	Quotation	Support line-item quotations, revisions, alternatives and locked submission.	Must	Submitted revisions are immutable and timestamped.
FR-018	PO	Expose PO terms and permit acknowledgement/proposed dates.	Must	External proposals do not overwrite PO data.

ID	Area	Requirement	Priority	Acceptance signal
FR-019	Delivery	Support dispatch, shipment and delivery evidence submissions.	Must	Inventory owns receipt and acceptance.
FR-020	Submittal	Support material, technical, drawing, method and certificate submissions.	Must	Revision and decision history is complete.
FR-021	Claims	Support external progress claims and supporting quantities/evidence.	Must	Claimed and certified values remain distinct.
FR-022	Invoices	Allow supplier invoice submission and status visibility.	Must	Posted invoice cannot be altered externally.
FR-023	QA/QC	Expose assigned NCRs, defects and corrective actions.	Must	QA/QC owns closure.
FR-024	HSE	Expose required HSE documents, actions and eligibility summaries.	Must	Restricted incident information is redacted.
FR-025	Documents	Provide controlled document rooms, uploads, downloads and transmittals.	Must	Documents owns file/version truth.
FR-026	Acknowledgement	Record acknowledgement by user, organization and document/record version.	Must	Read and acknowledgement remain distinct.
FR-027	Messaging	Provide contextual threads tied to business records.	Must	No unrestricted cross-tenant chat.
FR-028	Search	Search only externally published projections and permitted metadata.	Must	No internal or other-party index leakage.
FR-029	BIM	Show approved model versions, layers and object metadata.	Should	No unrestricted source model download.
FR-030	Mobile	Support core client and supplier workflows on mobile.	Must	Critical flows pass supported-device testing.
FR-031	Offline	Allow configured offline drafts with server-confirmed submission.	Should	No duplicate submission after reconnect.
FR-032	Branding	Support company-specific portal branding and terminology.	Should	Branding cannot change security behavior.
FR-033	Audit	Audit identity, access, views, downloads, submissions and denials.	Must	Sensitive actions are searchable by authorized admins.
FR-034	Notifications	Publish portal events to the Notifications & Event Centre.	Must	Portal does not implement channel logic independently.
FR-035	Configuration	Configure every portal page, widget, field and action.	Must	Disabled elements are removed from UI and API capability.

11. External Identity and Account Lifecycle

Identified -> Invited -> Invitation Opened -> Identity Verified
 -> Account Created -> Terms Accepted -> Membership Activated
 -> Business Access Granted -> Active

11.1 Alternative states

- Invitation Expired
- Invitation Cancelled
- Account Locked
- Membership Suspended
- Access Expired
- Relationship Ended
- Account Deactivated
- Archived

11.2 Separation of states

State	Meaning
Account state	Whether the person may authenticate.
Membership state	Whether the person represents a specified external organization.
Relationship state	Whether the membership/organization may access a specified business scope.
Publication state	Whether an internal record is currently externally visible.

A portal account may be active while all business relationships are expired. Authentication success must not imply access to projects, contracts, units or packages.

12. Invitations, Activation and Onboarding

12.1 Invitation types

- Individual invitation
- Organization administrator invitation
- Project invitation
- Unit-owner invitation
- Procurement/RFQ invitation
- Contract invitation
- Temporary document-room invitation
- Bulk invitation

12.2 Required invitation fields

- Recipient email
- External organization
- Portal type
- Inviting company/user
- Relationship and default role
- Allowed scopes
- Expiry
- Single-use token
- Required verification
- Terms/privacy version
- Status

12.3 Activation workflow

1. Validate token, expiry, intended email and invitation state.
2. Detect existing portal account and duplicate external-person candidates.
3. Verify contact method and identity according to policy.
4. Accept current terms, privacy notice and organization representation statement.
5. Create or attach portal membership.
6. Activate only the relationship scopes contained in the invitation.
7. Notify the inviter and record the activation audit event.

12.4 Safeguards

- No business data in invitation email
- Resend throttling
- Revocation
- Single-use token
- Domain allow/deny rules
- Existing-account linking
- Duplicate-person review
- No automatic company-wide access

13. External Organizations and Membership

13.1 External organization fields

- Legal/trading name
- Registration and tax identifiers
- Country and address
- Organization type
- Internal CRM/supplier reference
- Status
- Primary contacts
- Brand-safe display name

13.2 Membership fields

- External person
- External organization
- Portal role
- Job title
- Start/end date
- Primary contact flag
- Invitation source
- Status
- Delegated member-management capability

13.3 Organization administrator controls

- Invite members only within delegated organization and relationship scope.
- Suspend or remove own organization members where policy permits.
- Request additional project/package access but never grant it directly.
- View organization-level portal activity reports only where configured.

14. Business Access Relationships

14.1 Supported relationship types

Relationship	Example scope
Client organization to project	Project A client representatives.

Relationship	Example scope
Person to unit	Buyer or co-owner of Unit B-1204.
Supplier to RFQ	Invitation to procurement package RFQ-033.
Supplier to purchase order	PO and delivery collaboration.
Subcontractor to contract	Contract, claims, QA/QC and HSE workspace.
Consultant to document room	Temporary technical review workspace.
Client to financial account	Only invoices/statements for configured contract/unit.

14.2 Access evaluation

```

Allow = Active Account
      AND Active Membership
      AND Active Business Relationship
      AND Portal Role Capability
      AND Source Record Published
      AND Tenant/Company/Project/Record Scope Match
      AND Field/Document Classification Permits Access

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14.3 Revocation behavior

- Invalidate access tokens/caches
- Remove search entries
- Redact open previews
- Block new downloads
- Preserve audit and previously recorded acknowledgement
- Do not delete source records or submissions

15. Authentication, Sessions and Sensitive Actions

15.1 Authentication requirements

- Email/password or passwordless option
- Configurable MFA
- Device/session history
- Login alerts
- Brute-force and credential-stuffing protection
- Account lockout
- Recovery codes
- CAPTCHA/risk challenge where configured

15.2 Sensitive-action reauthentication

Action	Required control
Accept contract/PO where legally enabled	Recent MFA or equivalent strong reauthentication.
Change bank details	Reauthentication, out-of-band verification and internal Finance acceptance.
Submit high-value claim/invoice	Reauthentication and confirmation summary.
Download highly restricted package	Reauthentication, watermark and audit.
Manage organization administrators	Reauthentication and delegated-scope validation.
Change legal organization details	Reauthentication and internal validation.

16. Shared Portal Shell and Common Navigation

16.1 Common navigation

- Home
- My Projects
- My Contracts
- Documents
- Messages
- Notifications
- Requests / Submissions
- Calendar
- Help & Support
- Organization
- My Profile
- Security Settings

16.2 Shell requirements

- Navigation generated from active relationships and capabilities.
- Persistent organization/context switcher with clear external organization identity.
- Global unread/required-action indicators without exposing unauthorized record titles.
- Support contact and legal notices visible from every page.
- No links or search access to internal platform routes.

17. Client Portal Information Architecture

Section	Purpose
Client Dashboard	Portfolio-level summary and required actions.
Projects	Approved project overview and progress.
Units / Properties	Assigned unit details, selections, handover and warranty.
Progress	Published milestones, photos and approved progress.
Contracts	Issued contract summaries, amendments and notices.
Financials	Invoices, receipts, schedules and balances.
Documents	Approved files, transmittals and required reading.
Meetings	Invitations, agendas, minutes and decisions.
Requests	Client inquiries, changes, document and meeting requests.
Inspections & Handover	Scheduling, checklists, observations and sign-off.
Defects & Warranty	Defect and after-sales case submission/tracking.
BIM / 3D	Approved model-based view of project/unit.

18. Client Dashboard and Attention Centre

18.1 Dashboard widgets

- Active projects
- Project progress
- Upcoming milestones

- Unit status
- Contract summary
- Amount invoiced/paid/outstanding
- Documents requiring acknowledgement
- Upcoming meetings
- Open client requests
- Handover readiness
- Open defects
- Warranty cases
- Recent announcements

18.2 Attention rules

- Payment due
- Client decision required
- Selection deadline
- Document acknowledgement required
- Inspection scheduled
- Defect response received
- Contract amendment published
- Warranty case updated

18.3 Drill-down

Every dashboard value opens the permitted source projection or a filtered external list. The dashboard must not calculate independent contractual, financial or progress truth.

19. Client Project Workspace

Tab	Contents
Overview	Project identity, approved description, contacts, key dates and published status.
Progress	Approved progress, phases, milestones, images and client-safe notices.
Programme	Published milestone dates and approved changes.
Financial summary	Permitted contract/invoice/payment summary.
Contracts	Issued agreements, amendments, notices and client obligations.
Documents	Approved project document rooms and transmittals.
Meetings	Client-visible meetings, minutes and decisions.
Requests	Client submissions and responses.
Inspections / Handover	Scheduled inspections, checklists and handover package.
Defects / Warranty	Client-safe defect and warranty lifecycle.
BIM/3D	Approved model and object context.
Timeline	Published external activity only.

19.1 Explicitly hidden content

- Internal progress drafts
- Unaccepted work
- Supplier pricing and quotations
- Internal delay analysis
- Internal NCR discussion
- Employee information

- Internal forecasts and risk strategy
- Internal approval comments

20. Client Progress and Milestone Projection

20.1 Progress states

State	Meaning
Recorded	Captured internally by Work Progress.
Internally reviewed	Reviewed but not necessarily external-ready.
Client-visible	Explicitly published to the client.
Certified	Accepted through the relevant commercial/quality process.

20.2 Client-visible content

- Overall and phase progress
- Building/floor/unit progress where permitted
- Approved milestones
- Approved photographs/videos
- Planned completion date
- Client-safe delay notices
- Freshness timestamp and source

20.3 Rule

Progress rule: Work Progress owns construction progress. The portal displays an approved external projection and must clearly distinguish recorded, accepted and certified states.

21. Unit and Property Workspace

21.1 Unit fields

- Unit reference
- Project/building/floor/type
- Areas
- Commercial status
- Buyer relationship
- Contract status
- Payment plan
- Finish package
- Selections
- Approved variations
- Handover status
- Warranty status

21.2 Buyer actions

- View unit and documents
- Review payment plan
- Confirm finish selections
- Submit variation request
- Schedule inspection
- Submit defect

- Submit warranty request
- Acknowledge handover documents

21.3 Isolation

A buyer or co-owner may access only explicitly assigned units and records. Counts, search suggestions, URLs and 3D layers must not reveal unrelated units or buyer identities.

22. Client Financial Workspace

22.1 Permitted records

- Contract value where configured
- Payment schedule
- Issued invoices
- Credit notes
- Receipts
- Outstanding balance
- Due dates
- Approved variation values
- Client statement

22.2 Client actions

- Upload proof of payment
- Raise invoice query
- Request statement
- Download invoice/receipt
- Acknowledge payment-plan update

22.3 Security

- No journals, bank reconciliation, internal notes, margins or other clients.
- Payment proof is a submission until Finance validates it.
- Any bank/payment instructions shown must be sourced from controlled Finance data and versioned.

23. Client Contract Workspace

23.1 Client-visible contract content

- Reference and parties
- Effective date
- Scope summary
- Contract value where allowed
- Payment schedule
- Key dates
- Approved amendments
- Status
- Issued documents
- Notices
- Client obligations
- Acknowledgement history

23.2 Client actions

- View/download
- Acknowledge
- Submit required information
- Respond to notice
- Request clarification

- Accept only where legally configured

23.3 Revision rule

Issued contract documents and notices are immutable. Corrections or changes create a new issued revision; the portal preserves which version each recipient viewed or acknowledged.

24. Client Requests, Decisions and Variations

24.1 Request types

- General inquiry
- Design clarification
- Variation request
- Payment inquiry
- Document request
- Meeting request
- Unit selection
- Inspection request
- Complaint
- Other configurable type

24.2 Lifecycle

Draft -> Submitted -> Received -> Under Review
-> Information Required -> Response Issued -> Resolved -> Closed

24.3 Decision requests

- Internal module publishes a decision request with choices, context, deadline and consequences.
- Client submits a response or requests clarification.
- Response is immutable and does not update the source record until internal processing completes.
- Workflow orchestration will later be standardized by the final Approvals & Workflow Engine.

25. Client Inspections, Handover and Sign-Off

25.1 Inspection record

- Project/unit
- Inspection type
- Scheduled date/time
- Participants
- Checklist
- Observations
- Attachments
- Client comments
- Sign-off/acknowledgement
- Status

25.2 Handover package

- Handover readiness summary
- Issued documents
- Keys/assets list where applicable
- Meter readings
- Training/operation information
- Open accepted defects
- Warranty dates

- Client acknowledgement

25.3 Sign-off distinction

Portal acknowledgement, inspection attendance, acceptance of a handover checklist and formal legal handover acceptance are separate actions with separate capabilities and evidence.

26. Defects, Warranty and After-Sales

26.1 Defect submission fields

- Unit/location
- Category
- Description
- Photo/video
- Client-reported severity
- Preferred access time
- Date reported
- Contact
- Internal QA/QC reference
- Status

26.2 Defect lifecycle

Submitted -> Validated -> Accepted for Action -> Scheduled
-> Work Completed -> Client Review -> Closed

26.3 Warranty case

- Coverage check
- Inspection
- Responsible contractor projection
- Resolution evidence
- Client review
- Closure
- Reopen reason

Internal cost recovery, liability allocation, contractor disputes and legal strategy remain hidden.

27. Supplier Portal Information Architecture

Section	Purpose
Supplier Dashboard	Commercial, delivery, compliance and action summary.
Company Profile	Legal identity, contacts, categories and capacity.
Prequalification	Questionnaires, evidence and status.
Opportunities / RFQs	Invitations, clarifications, addenda and deadlines.
Quotations	Draft, revise, submit and track bid status.
Purchase Orders	Issued terms, acknowledgements and delivery commitments.
Submittals	Material, technical, drawing and method submissions.
Deliveries	Dispatch, shipment and receiving collaboration.
Progress Claims	Quantities, evidence, certification and payment status.
Timesheets	External labour/crew submissions where applicable.

Section	Purpose
Invoices / Payments	Submit invoice and view validation/payment status.
QA/QC	NCR, defect and corrective-action collaboration.
HSE	Requirements, documents, observations and actions.

28. Supplier Dashboard and Attention Centre

28.1 Dashboard widgets

- Open RFQs
- Quotation deadlines
- Active POs
- Delivery commitments
- Submittals awaiting response
- Progress claims
- Invoice/payment status
- Compliance expirations
- HSE requirements
- QA/QC actions
- Meetings
- Unread messages

28.2 Attention centre

- RFQ closing soon
- PO acknowledgement required
- Delivery overdue
- Submission returned
- Insurance expiring
- HSE document missing
- Claim returned
- Invoice rejected
- Corrective action overdue
- Contract notice received

29. Supplier Company Profile and Prequalification

29.1 Supplier profile

- Legal identity
- Registration/tax data
- Addresses
- Contacts
- Service categories
- Regions
- Certifications
- Insurance
- Licences
- HSE/quality systems
- References
- Capacity
- Equipment/workforce summary
- Bank-detail status

29.2 Update model

Supplier proposes profile update
 -> Profile Update Submission
 -> Internal validation
 -> Accepted fields update Supplier master record

29.3 Prequalification lifecycle

Not Started -> In Progress -> Submitted -> Under Review
 -> Clarification Required -> Qualified / Conditionally Qualified / Not Qualified -> Expired

30. Supplier Compliance Documents

30.1 Common requirements

- Company registration
- Tax certificate
- Insurance
- HSE policy
- Quality certificate
- Licences
- Authorized signatory
- Bank confirmation
- Workforce certificates
- Product certificates
- Environmental documentation

30.2 Requirement fields

- Document type
- Issuer
- Issue/expiry date
- Required scope
- Uploaded revision
- Validation status
- Reviewer comments
- Renewal reminder
- Related project/contract

30.3 Expiry behavior

- Warn supplier and internal owner
- Show affected relationships
- Do not silently extend validity
- Optionally restrict new submissions or site eligibility according to source-module policy

31. RFQ and Tender Workspace

31.1 RFQ content

- RFQ reference
- Package/project
- Scope
- Commercial/technical requirements
- Dates
- Clarifications

- Addenda
- Documents
- Submission rules
- Status

31.2 Supplier actions

- Accept/decline invitation
- Ask clarification
- Acknowledge addendum
- Prepare quotation
- Upload evidence
- Submit/withdraw before deadline where allowed

31.3 Confidentiality

Tender confidentiality: A supplier must never see competing suppliers, other quotations, internal target prices, internal evaluations, negotiation notes or award strategy.

32. Quotation and Revision Management

32.1 Quotation fields

- Currency
- Validity
- Lead time
- Delivery/payment terms
- Line-item prices
- Tax
- Discount
- Exclusions
- Alternatives
- Technical/commercial attachments
- Total
- Revision
- Status

32.2 Lifecycle

Draft -> Submitted -> Submission Locked -> Clarification
-> Revised Submission -> Evaluation Complete -> Awarded / Not Awarded

32.3 Rules

- Submitted revision is immutable.
- Resubmission creates a new revision.
- Deadline enforcement uses server time and configured time zone.
- Late submission is blocked or flagged according to RFQ policy.
- Award outcome may be published without exposing evaluation detail.

33. Purchase Orders and Change Communication

33.1 Supplier-visible PO data

- Reference
- Project/buyer

- Scope
- Line items
- Quantity/price
- Delivery location/dates
- Payment terms
- Special conditions
- Documents
- Change orders
- Status

33.2 Supplier actions

- Acknowledge
- Accept where configured
- Request clarification
- Propose delivery schedule
- Submit dispatch notice
- Upload delivery documents
- Raise change request

33.3 Change rule

Supplier acknowledgement confirms receipt of the issued version. A proposed change is an external submission and does not modify the Purchase Order until Procurement accepts and issues a new version/change order.

34. Delivery and Receiving Collaboration

34.1 Delivery record

- PO/shipment
- Planned/confirmed/dispatch/arrival dates
- Delivery location
- Items/quantities
- Transport
- Packing list
- Certificates
- Photos
- Status

34.2 Lifecycle

Planned -> Supplier Confirmed -> Dispatched -> In Transit
-> Arrived -> Receiving Inspection -> Accepted / Partially Accepted / Rejected -> Closed

34.3 Ownership

The supplier reports dispatch and shipment information. Inventory owns receipt, accepted quantities and stock movement. QA/QC may own receiving inspection outcomes.

35. Material, Technical and Design Submissions

35.1 Submission types

- Material approval
- Product data
- Shop drawing
- Method statement
- Sample/mock-up

- Certificate/test report
- Technical query
- Design/replacement proposal

35.2 Submission lifecycle

Draft -> Submitted -> Technical Review -> Clarification Required
-> Resubmission -> Accepted / Accepted with Comments / Rejected -> Superseded

35.3 Revision controls

- Submission number and revision are immutable after submit.
- Files are tied to the exact revision.
- Internal comments are separated from published response comments.
- Superseded revisions remain accessible to authorized parties.

36. Supplier Contract, Notice and Obligation Workspace

36.1 Supplier-visible content

- Contract summary
- Scope
- Commercial terms
- Key dates
- Bonds/insurance
- Deliverables
- Notices
- Change orders
- Payment terms
- Compliance obligations
- Documents

36.2 Supplier actions

- Acknowledge
- Submit deliverable
- Respond to notice
- Request clarification
- Submit change proposal
- Upload insurance/guarantee

36.3 Notices

Notices are issued records with recipient, date, version, response requirement and legal classification. Messaging must not substitute for a formal notice where the Contracts or Government & Legal module requires one.

37. Progress Claims and External Timesheets

37.1 Claim fields

- Contract/period
- Work items
- Previous/current/cumulative quantities
- Retention
- Advance recovery
- Materials on site
- Variations
- Deductions

- Tax
- Claimed amount
- Evidence

37.2 Lifecycle

Draft -> Submitted -> Under Verification -> Clarification Required
 -> Revised -> Certified / Partially Certified / Rejected
 -> Finance Processing -> Paid

37.3 Timesheets

- Worker/crew label
- Date
- Project/location/work package
- Role
- Hours/overtime
- Supervisor
- Notes
- Evidence
- Verification status

Commercial rule: Supplier-submitted quantities and hours are claims. Work Progress, QA/QC, Contracts and Finance determine accepted, certified and paid values.

38. Supplier Invoices and Payment Visibility

38.1 Invoice submission

- Invoice number/date
- PO/contract/claim/delivery reference
- Currency/tax
- Line items
- Bank-detail reference
- Attachments
- Total

38.2 External status

- Submitted
- Validation Failed
- Under Review
- Accepted
- Disputed
- Posted
- Payment Scheduled
- Partially Paid
- Paid
- Rejected

38.3 Rules

- Posted invoices cannot be edited externally.
- Correction requires a new document/credit note process.
- Payment status is a Finance projection and includes freshness.
- Internal holds, fraud indicators and bank data are not exposed unless explicitly published.

39. Supplier QA/QC and Corrective Actions

39.1 Externally visible records

- Inspection requests
- Material inspections
- Assigned NCRs
- Defects
- Corrective actions
- Tests
- Reinspection results
- Closure evidence

39.2 Corrective-action lifecycle

Issued -> Supplier Response -> Corrective Plan Submitted
-> Work in Progress -> Evidence Submitted -> Internal Verification -> Closed / Returned

39.3 Rule

Supplier evidence and proposed closure are submissions. QA/QC owns formal acceptance, status and closure.

40. Supplier HSE and Site Eligibility

40.1 Supplier HSE workspace

- Project HSE requirements
- Mandatory documents
- Worker eligibility summaries
- Inductions
- Toolbox talks
- Permit requirements
- Inspections
- Corrective actions
- Safety announcements

40.2 Supplier reports

- Near miss
- Hazard
- Incident
- Unsafe condition
- Corrective-action response

40.3 Privacy

The portal exposes only the safety information needed for the supplier relationship. Sensitive medical, witness, investigation or employee information remains protected in HSE/HR.

41. Meetings, Messaging and Support

41.1 Portal meetings

- View invitation
- Accept/decline
- View agenda

- Upload requested material
- Join through configured link
- View issued minutes
- Acknowledge decision
- Request correction

41.2 Contextual messaging

- Project
- Contract
- RFQ
- PO
- Submission
- Claim
- Defect
- Warranty case
- Invoice

Messaging is contextual and auditable. It must not become an unrestricted general chat system or replace issued documents/notices.

41.3 Support cases

- Authentication/access issue
- Portal navigation issue
- Document issue
- Submission issue
- Billing question
- Security report
- Other configurable category

42. Document Rooms, Transmittals and Controlled Exchange

42.1 External document room

- Audience
- Scope
- Folder structure
- Allowed actions
- Effective dates
- Watermark/download policy
- Expiry
- Activity log

42.2 Document classifications

- External Shareable
- Client Restricted
- Supplier Restricted
- Contract Party Only
- Project External
- Internal Only
- Confidential
- Highly Restricted

42.3 Transmittal lifecycle

Draft -> Issued -> Delivered -> Opened -> Acknowledged -> Responded -> Closed

42.4 Rules

- Documents owns files and revisions.

- Folder inheritance alone cannot publish internal-only content.
- Each recipient has separate delivery/open/acknowledgement history.
- Downloaded content may be watermarked and use short-lived links.

43. BIM/3D Portal Viewer

43.1 Client use cases

- View approved project/unit model
- View approved design/progress
- Navigate handover locations
- View permitted defects linked to elements

43.2 Supplier use cases

- View relevant scope zones/elements
- Open linked issued drawings
- Submit model-related issue
- View inspection/corrective-action location

43.3 Restrictions

- Approved versions only
- Allowed layers/elements only
- No unrestricted model/source download
- No hidden commercial metadata
- No unrelated units/scopes
- No internal markups
- Permission-aware properties
- Watermark/session tracking where required

44. Notifications, Acknowledgements and Required Reading

44.1 Portal events

- Invitation received
- New RFQ/addendum
- PO issued
- Submission returned
- Delivery changed
- Invoice/payment status changed
- Document published
- Contract notice
- Client decision required
- Inspection scheduled
- Defect/warranty updated
- Compliance expiring
- Emergency HSE notice

44.2 Acknowledgement evidence

- User
- Organization
- Record/document revision
- Timestamp
- Confirmation statement
- Session/device
- Optional comment
- Event version

44.3 Distinctions

Action	Meaning
Read	User opened the notification or record.
Acknowledge	User confirmed receipt/awareness of a specific version.
Accept	User performed an explicitly enabled legal/commercial action.
Approve	Internal decision governed by source module/future workflow engine.

45. External Submission and Revision Model

45.1 Standard envelope

```

submission_id
submission_type
portal_type
external_organization_id
submitted_by
tenant_id / company_id / project_id
source_context_type / source_context_id
revision
submitted_at
status
payload
document_references
correlation_id
idempotency_key

```

45.2 Common lifecycle

```

Draft -> Validated -> Submitted -> Received -> Under Review
-> Clarification Required -> Resubmitted
-> Accepted / Partially Accepted / Rejected -> Closed

```

45.3 Submission principles

- Submitted revision is immutable.
- Resubmission creates a new revision.
- Withdrawal is controlled and audited.
- Internal and external comments are separate.
- Source-module conversion occurs only after internal acceptance.
- Duplicate requests are prevented by idempotency keys and business rules.

46. Projection and Publication Architecture

46.1 Projection flow

```

Source Module
-> External Publication Policy
-> Permission-Safe Projection Builder
-> Versioned Portal Projection
-> Portal API / Search Index / Cache
-> Authorized External User

```

46.2 Projection fields

- Source entity/version
- External wording
- Visible fields
- Audience/scope
- Published version
- Published at/by
- Expiry
- Sensitivity
- Deep links/actions
- Freshness

46.3 Publication lifecycle

Draft -> Ready for Publication -> Published -> Updated
-> Withdrawn / Expired -> Archived

46.4 Cache invalidation

- Source event changes
- Permission/relationship change
- Publication withdrawal
- Document classification change
- User/membership suspension
- Retention/legal-hold action

47. Data Model

- ExternalPerson, ExternalOrganization, PortalAccount, PortalMembership, PortalRole, PortalCapability
- PortalInvitation, BusinessAccessRelationship, PortalAccessGrant, PortalAccessScope, PortalSession, PortalDevice
- PortalProjection, ProjectionVersion, ExternalSubmission, SubmissionRevision, SubmissionComment, PortalRequest
- PortalMessageThread, PortalMessage, ExternalDocumentRoom, DocumentRoomMembership, PortalTransmittal, TransmittalRecipient
- ExternalAcknowledgement, ClientProjectProjection, ClientUnitRelationship, SupplierQualificationProjection, SupplierOpportunityProjection, SupplierContractProjection
- SupplierDeliverySubmission, SupplierClaimSubmission, SupplierInvoiceSubmission, ExternalInspectionParticipation, PortalNotificationPreference, PortalBrandConfiguration
- PortalSupportCase, PortalActivity

47.1 Common entity requirements

- tenant_id
- company_id where applicable
- record_version
- created/updated timestamps
- created/updated by
- effective_from/effective_to
- status
- confidentiality/sensitivity
- correlation_id
- audit metadata

47.2 Projection requirements

- source_entity_type/id/version
- publication_status
- published_version
- published_at/by
- audience_scope
- expires_at

- visible-field schema version
- checksum

48. Backend Services and Domain Events

48.1 Recommended services

- External Identity Service
- Portal Account Service
- Invitation Service
- External Organization Service
- Membership Service
- Portal Access Policy Service
- Relationship Resolution Service
- Projection Service
- External Submission Service
- Client Portal Service
- Supplier Portal Service
- Document Room Service
- Transmittal Service
- Messaging Service
- Acknowledgement Service
- Notification Adapter
- Portal Search Service
- Branding Service
- Support Service
- Audit Service
- Source Module Adapters
- Malware Scanning/File Validation Service

48.2 Source-to-portal events

- ProjectClientAccessGranted/Revoked
- UnitBuyerAssigned
- ClientDocumentPublished
- SupplierInvitedToRFQ
- RFQAddendumIssued
- PurchaseOrderIssued
- DeliveryStatusChanged
- SubmissionDecisionIssued
- InvoiceStatusChanged
- PaymentStatusChanged
- DefectStatusChanged
- HandoverScheduled
- WarrantyCaseUpdated
- SupplierComplianceExpiring

48.3 Portal-to-source events

- PortalInvitationAccepted
- PortalUserActivated
- ExternalSubmissionReceived
- ExternalDocumentDownloaded
- ExternalAcknowledgementRecorded
- ClientRequestSubmitted
- SupplierQuotationSubmitted
- SupplierClaimSubmitted
- PortalAccessRevoked

48.4 Processing requirements

- Transactional outbox

- Idempotent consumers
- Schema versioning
- Retry queues
- Dead-letter handling
- Correlation/causation IDs
- No direct cross-module database writes

49. API Contract

49.1 Identity and organization

```
POST /portal/invitations/accept
POST /portal/accounts
POST /portal/auth/login
POST /portal/auth/mfa/verify
GET /portal/me
GET /portal/organization
GET /portal/organization/members
POST /portal/organization/invitations
PATCH /portal/organization/members/{id}
```

49.2 Client APIs

```
GET /portal/client/projects
GET /portal/client/projects/{id}
GET /portal/client/units
GET /portal/client/units/{id}
GET /portal/client/contracts
GET /portal/client/financials
POST /portal/client/requests
POST /portal/client/defects
POST /portal/client/warranty-cases
POST /portal/client/inspections/{id}/acknowledge
```

49.3 Supplier APIs

```
GET /portal/supplier/opportunities
GET /portal/supplier/rfqs/{id}
POST /portal/supplier/rfqs/{id}/quotations
POST /portal/supplier/quotations/{id}/submit
GET /portal/supplier/purchase-orders
POST /portal/supplier/purchase-orders/{id}/acknowledge
POST /portal/supplier/deliveries
POST /portal/supplier/submittals
POST /portal/supplier/progress-claims
POST /portal/supplier/invoices
```

49.4 API requirements

- Idempotency key on writes
- Active account/membership/relationship validation
- Record version on mutable drafts
- Server-side field allowlist
- File quarantine token where applicable
- Audit metadata
- Cursor pagination
- Rate limiting
- No unrestricted generic source-record endpoint

50. Frontend Architecture and UX States

50.1 Shared components

- PortalShell
- PortalHeader
- ContextSwitcher
- PortalNavigation
- PortalDashboard
- PortalNotificationCentre
- PortalDocumentViewer
- PortalUploadPanel
- PortalSubmissionWizard
- PortalStatusTimeline
- PortalMessageThread
- PortalAcknowledgementModal
- PortalProfilePage
- PortalOrganizationPage
- PortalSecurityPage
- PortalSupportPage

50.2 Client components

- ClientProjectCard/Page
- ClientUnitPage
- ClientProgressView
- ClientFinancialSummary
- ClientContractView
- ClientRequestForm
- ClientInspectionPage
- ClientDefectForm
- ClientWarrantyPage
- ClientBIMViewer

50.3 Supplier components

- SupplierDashboard
- SupplierProfileEditor
- PrequalificationWizard
- RFQWorkspace
- QuotationBuilder
- PurchaseOrderPage
- DeliveryScheduler
- SubmittalWorkspace
- ProgressClaimBuilder
- SupplierInvoiceForm
- SupplierQACorrectiveActionPage
- SupplierHSEWorkspace

50.4 Required UX states

- Loading/skeleton
- Empty but authorized
- No relationship/access
- Expired relationship
- Submission draft
- Validation errors
- Submitted/locked
- Clarification required
- Permission revoked while open
- Offline draft
- File quarantined

- Service temporarily unavailable

51. Search, File Security and Malware Controls

51.1 Search scope

- Published projects/units/contracts
- RFQs/POs
- Documents
- Submissions/claims/invoices
- Defects/warranty
- Messages

51.2 Search exclusions

- Internal comments
- Hidden fields
- Unpublished records
- Other organizations' records
- Internal audit data
- Restricted filenames/metadata

51.3 Upload controls

- File type/size validation
- Malware scanning
- Quarantine
- Checksum
- Duplicate detection
- Metadata cleaning where required
- Content policy validation
- Safe preview generation

51.4 Download controls

- Short-lived signed URL
- Watermark
- User/organization/timestamp stamp
- View-only mode
- Download restriction
- Audit
- Revocation check before issue

52. Branding, Localization and Accessibility

52.1 Branding

- Portal name/logo/colors/domain
- Login image
- Support contact
- Footer/legal text
- Terms/privacy
- Email templates
- Watermarks
- Client/supplier terminology

52.2 Localization

- Multiple languages
- Time zones

- Locale dates/numbers
- Currencies
- Translated templates
- Right-to-left readiness
- Original-language preservation for issued documents

52.3 Accessibility

- WCAG-aligned keyboard navigation
- Visible focus
- Screen-reader labels
- High contrast
- No color-only status
- Reduced motion
- Accessible tables/forms/errors
- Caption/alternative content for essential media where available

53. Responsive, Mobile and Offline Behavior

53.1 Client mobile priorities

- Project/unit summary
- Payment status
- Documents
- Requests
- Defects/warranty
- Notifications
- Inspection schedule
- Simplified 3D where supported

53.2 Supplier mobile priorities

- RFQ alerts
- PO acknowledgement
- Delivery updates
- Uploads
- Site submissions
- Timesheets
- QA/HSE evidence
- Invoice/payment status

53.3 Offline drafts

- Client defect/warranty draft
- Supplier delivery draft
- Timesheet draft
- QA evidence
- HSE observation
- Photo/document upload queue

53.4 Offline rules

- Formal submission requires server confirmation.
- Drafts are encrypted and scoped to account/context.
- Permission/relationship rechecked on reconnect.
- Duplicate submissions prevented through idempotency.
- Expired access blocks synchronization and clears protected cache according to policy.

54. Security, Privacy, Legal Hold and Audit

54.1 Security controls

- Tenant/organization/project/record isolation
- Field-level redaction
- MFA and reauthentication
- Session/device controls
- Rate limiting and anti-automation
- Encrypted transport/storage
- Secrets management
- Secure headers/CSP
- No sensitive data in logs
- Export/download audit

54.2 Privacy

- Data minimization
- Purpose limitation
- Terms/consent version history
- Right-to-access/correction workflows where required
- Controlled deletion/anonymization
- No inference through counts, filenames or URLs

54.3 Legal hold and retention

- Legal hold blocks deletion of submissions, issued documents, acknowledgements and audit.
- Retention differs by record/document type and legal company.
- Account deletion does not erase legally required business evidence.

54.4 Audit events

- Invitation create/accept
- Login/MFA change
- Membership/access grant/revoke
- Record view
- Restricted download
- Submission create/revise/withdraw
- Message
- Acknowledgement
- Member suspension
- Bank detail proposal
- File-security rejection
- Permission denied

55. Administration and Platform Configuration

55.1 Administrative areas

- Portal types
- Branding
- External roles/capabilities
- Invitation templates/expiry
- MFA/session policy
- Relationship types
- Publication policies
- Visible-field schemas
- Document classifications
- Document-room templates
- Submission types/statuses
- Request categories
- Acknowledgement policy
- File/download/watermark rules
- Compliance requirements

- Dashboard widgets
- Localization
- Retention
- Support categories
- Notification hooks
- Approval hooks
- Custom fields

55.2 Configuration principle

Every portal page, section, widget, field, action and export must be configurable. Configuration controls availability and presentation; permissions independently determine whether the current user may access the configured capability.

56. Reporting, Analytics and Operational Monitoring

56.1 Internal portal analytics

- Active users
- Invitation activation
- Login success/failure
- Document views/downloads
- Required acknowledgements
- Submission volume/response time
- Open client requests
- RFQ participation
- Compliance completion
- Support cases
- Access-denied attempts

56.2 Operational metrics

- API latency/error rate
- Projection refresh lag
- Event queue lag
- Search indexing lag
- File-scan queue and failures
- Upload/download throughput
- Session/MFA failures
- Notification delivery failures
- Cache hit rate

56.3 Guardrail

Analytics supports operations and service improvement. It must not automatically become contractual supplier scoring, employee performance evaluation or client-risk classification.

57. Non-Functional Requirements

57.1 Performance targets

Area	Target behavior
Dashboard	Cached primary dashboard loads within approximately 3 seconds under normal broadband.
Record view	Common published record view responds within approximately 2 seconds.
Lists	Cursor pagination and virtualized tables for large result sets.

Area	Target behavior
Uploads	Resumable upload for large packages where practical; background scanning with progress.
3D	Progressive streaming and device-specific quality profile.
Submissions	Server-confirmed write with idempotent retry.
Exports	Asynchronous for large outputs with notification on completion.

57.2 Availability and resilience

- Graceful read-only behavior for cached safe projections during temporary source outage.
- Submission queue preserves user input when downstream source adapter is unavailable.
- One provider/module failure does not take down unrelated portal areas.
- Retry-safe events and dead-letter handling.
- Defined RPO/RTO and backup restoration tests.

57.3 Scalability

- Thousands of concurrent external users
- Large tender events
- Large document packages
- High-volume photo uploads
- Multi-company branding/domains
- Simultaneous announcements
- Independent horizontal scaling of projections, search, files and submissions

58. Testing Strategy and Critical Scenarios

58.1 Required test layers

- Unit tests for policy evaluation, relationship resolution and field allowlists.
- API contract tests for every source adapter.
- Integration tests for projection publication and submission conversion.
- Security tests for cross-organization and URL manipulation.
- File-security tests for malware/quarantine and signed downloads.
- Mobile/responsive/offline synchronization tests.
- Accessibility and localization tests.
- Load tests for tenders, announcements and file traffic.
- Audit completeness and legal-hold tests.

58.2 Critical scenarios

1. Supplier attempts to access another supplier's quotation.
2. Client loses project access while the portal page is open.
3. External user belongs to two client organizations and switches context.
4. Supplier administrator invites a user beyond delegated scope.
5. RFQ is cancelled after quotation submission.
6. PO changes after supplier acknowledgement.
7. Client submits the same defect twice while reconnecting from offline.
8. Supplier proposes new bank details before invoice payment.
9. Document access expires after a file was cached locally.
10. User is deactivated while submitting a high-value claim.
11. One client is linked to two units in different projects.
12. External organization changes legal identity.
13. Internal record is withdrawn after external download.
14. Source-module permissions change while a portal session is active.
15. Uploaded file passes extension validation but contains malware.

- 16. Projection event is delivered twice.
- 17. Published document revision is superseded after acknowledgement.
- 18. Client search attempts to enumerate unit references.

59. Acceptance Criteria

ID	Area	Acceptance criterion	Evidence
AC-001	Identity	External users authenticate through portal accounts and never receive internal employee navigation or permissions.	Pass when verified by automated or documented QA evidence.
AC-002	Invitation	Invitation scope, expiry and single-use behavior are enforced.	Pass when verified by automated or documented QA evidence.
AC-003	Membership	One person can represent multiple external organizations without data crossover.	Pass when verified by automated or documented QA evidence.
AC-004	Relationship	An active account without an active relationship sees no business records.	Pass when verified by automated or documented QA evidence.
AC-005	Projection	Portal APIs return only published, allowlisted fields.	Pass when verified by automated or documented QA evidence.
AC-006	Revocation	Revoked access is removed from open sessions, search, caches and downloads.	Pass when verified by automated or documented QA evidence.
AC-007	Client projects	Client sees only explicitly published project data.	Pass when verified by automated or documented QA evidence.
AC-008	Units	Buyer sees only assigned units and associated records.	Pass when verified by automated or documented QA evidence.
AC-009	Financials	Client/supplier financial views reconcile to source projections and hide internal accounting fields.	Pass when verified by automated or documented QA evidence.
AC-010	Requests	Client request submissions preserve immutable revisions and status history.	Pass when verified by automated or documented QA evidence.
AC-011	Defects	Defect/warranty submissions link to QA/QC without allowing external closure.	Pass when verified by automated or documented QA evidence.
AC-012	Supplier profile	Supplier profile changes require internal validation before master update.	Pass when verified by automated or documented QA evidence.
AC-013	Prequalification	Questionnaire, evidence, clarification and decision states are complete.	Pass when verified by automated or documented QA evidence.
AC-014	RFQ confidentiality	No supplier can infer competitors, quotations or internal evaluation.	Pass when verified by automated or documented QA evidence.
AC-015	Quotation	Submitted quotation revisions are locked, timestamped and immutable.	Pass when verified by automated or documented QA evidence.
AC-016	PO	Supplier acknowledgement references the exact issued PO version.	Pass when verified by automated or documented QA evidence.

ID	Area	Acceptance criterion	Evidence
AC-017	Delivery	Supplier dispatch updates do not create stock movements until Inventory receives them.	Pass when verified by automated or documented QA evidence.
AC-018	Submittals	Every resubmission creates a new revision with complete decision history.	Pass when verified by automated or documented QA evidence.
AC-019	Claims	Claimed, certified and paid amounts remain distinct.	Pass when verified by automated or documented QA evidence.
AC-020	Invoices	Posted invoices cannot be edited through the portal.	Pass when verified by automated or documented QA evidence.
AC-021	QA/QC	Supplier evidence cannot close an NCR/defect without QA/QC verification.	Pass when verified by automated or documented QA evidence.
AC-022	HSE	Supplier HSE view exposes only relationship-relevant data.	Pass when verified by automated or documented QA evidence.
AC-023	Documents	Internal-only files cannot become external through folder inheritance.	Pass when verified by automated or documented QA evidence.
AC-024	Transmittals	Each recipient has independent delivery/open/acknowledgement history.	Pass when verified by automated or documented QA evidence.
AC-025	Acknowledgement	Read, acknowledge, accept and approve are distinct actions.	Pass when verified by automated or documented QA evidence.
AC-026	BIM	Only approved model versions/layers/metadata are visible.	Pass when verified by automated or documented QA evidence.
AC-027	Files	All uploads are quarantined and scanned before acceptance.	Pass when verified by automated or documented QA evidence.
AC-028	Search	Search index contains only externally published projections.	Pass when verified by automated or documented QA evidence.
AC-029	Mobile	Critical client and supplier workflows function on supported mobile devices.	Pass when verified by automated or documented QA evidence.
AC-030	Offline	Offline drafts synchronize idempotently after permission recheck.	Pass when verified by automated or documented QA evidence.
AC-031	Branding	Company branding changes appearance only, not security.	Pass when verified by automated or documented QA evidence.
AC-032	Audit	Sensitive access, downloads, submissions and denials are fully auditable.	Pass when verified by automated or documented QA evidence.
AC-033	Configuration	Page, field and action configuration is enforced consistently in UI and APIs.	Pass when verified by automated or documented QA evidence.
AC-034	Integration	All source interactions use APIs/events; no direct database coupling.	Pass when verified by automated or documented QA evidence.

60. Delivery Phases and Migration

Phase	Scope
Phase 1 - Shared foundation	External identity, invitations, MFA, organizations, memberships, relationships, permissions, shared shell, documents, notifications, messages, audit and branding.
Phase 2 - Client Portal	Dashboard, projects, units, progress, contracts, financial summary, requests, meetings, inspections, defects and warranty.
Phase 3 - Supplier procurement	Supplier profile, prequalification, compliance, RFQs, quotations, POs, deliveries, submittals and documents.
Phase 4 - Supplier execution/commercial	Claims, timesheets, invoices, payment status, QA/QC, HSE, corrective actions and contract notices.
Phase 5 - Advanced external experience	Controlled BIM/3D, expanded offline, advanced analytics and future portal types.
Future integrations	E-signature, payment gateways, external identity providers, external storage and additional collaboration providers.

60.1 Migration inputs

- Existing client/supplier contacts
- CRM/supplier organization records
- Active projects/units/contracts/POs
- Existing portal/file-share access lists
- Compliance document registers
- Current external document packages

60.2 Migration controls

- Duplicate person/organization matching
- Relationship and scope validation
- Document classification review
- Pilot users/organizations
- Parallel access before cutover
- Invitation and support campaign
- Revocation of legacy sharing links after acceptance

61. Page and Flow Registry

Page / flow	Primary purpose
Portal Login / Activation	Secure account activation, authentication, MFA and recovery.
Context Switcher	Switch external organization and business context.
Client Dashboard	Client portfolio overview and required actions.
Client Project Page	Published project workspace.
Client Unit Page	Buyer/unit workspace.
Client Request Wizard	Create inquiries, variations and decisions.
Handover / Defect / Warranty	Inspection and after-sales collaboration.
Supplier Dashboard	Supplier commitments and required actions.
Supplier Profile / Prequalification	Organization data and qualification.
RFQ Workspace / Quotation Builder	Tender response.
PO / Delivery Page	Order acknowledgement and delivery collaboration.
Submittal Workspace	Material/technical submission lifecycle.

Page / flow	Primary purpose
Claim / Timesheet Builder	External commercial quantity/hour submission.
Supplier Invoice Form	Invoice submission and status.
QA/QC / HSE Workspace	Corrective actions, evidence and safety requirements.
Document Room / Transmittal	Controlled file exchange.
Portal Organization Admin	Manage organization members within scope.
Portal Administration	Policies, publication, branding, roles and monitoring.

62. Risks, Open Decisions and Definition of Done

62.1 Open decisions

Topic	Decision required
Portal domain model	Single shared domain with routes vs separate hostnames per portal/company.
Authentication	Password plus MFA baseline and passwordless availability.
Identity verification	Required level by portal role and sensitive action.
Client financial detail	Exact fields and statement formats by legal company.
Legal acceptance	Which portal actions constitute legal acceptance vs acknowledgement.
Offline scope	Supported draft types, cache duration and maximum file package.
BIM scope	Initial device/browser support and model download restrictions.
Organization administrators	Delegated invitation and suspension limits.
External sharing	Whether external users may share within their own organization.
Data residency	Storage/processing regions and company-specific restrictions.
Retention	Record-by-record retention and legal-hold policy.
Legacy migration	Which existing client/supplier access channels are retired.

62.2 Definition of Done

- All Must requirements and acceptance criteria are implemented and tested.
- External identity, membership and relationship boundaries are enforced in APIs, search, caches, files and UI.
- Client and supplier experiences are distinct and do not expose internal navigation or terminology unnecessarily.
- Every external write is an immutable submission/request/acknowledgement with complete audit history.
- Projection publication, withdrawal, expiry and permission revocation are production-ready.
- Cross-organization, cross-project, RFQ confidentiality and document-leakage tests pass.
- File quarantine, malware scanning, signed download and audit controls are operational.
- Mobile critical flows, accessibility and supported localization pass QA.
- Operational dashboards, alerts, runbooks, backups and recovery tests are approved.
- No unresolved critical security, privacy, contractual or data-isolation defect remains.

Final product position: The Client & Supplier Portals are purpose-built, relationship-based external workspaces. They expose only approved information and turn every external action into a secure, traceable submission while internal ConstructionOS modules remain the authoritative source of truth.